

Spatial misallocation under agglomeration externalities, isolating the causal effect of rent-sharing

MSc Economics dissertation proposal

Aloysius Lip^{1,2}

Supervisor: Lars Nesheim^{1,3}

¹Department of Economics, University College London (UCL)

²HM Treasury (HMT)

³Centre for Microdata Methods and Practice (cenmap)

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Core: an ML exercise process to estimate varied distance and firm characteristic effects on agglomeration. We want to measure strength and heterogeneity.

Analysis: we need valid causal inference, so draw on the following fields of research

Spatial/urban econometrics (applying learning from **Time series econometrics** course) as appropriate

Macroeconomic policy research learnt for identification of exogenous shocks

Extensions:

- (a) Clustering regressor (**ML in economics**)
- (b) Regularisation to remove ineffective characteristic interactions from neighbouring firms (**ML**)
- (c) Literature from social peer effects literature (**Labour economics**)
- (d) Introduce regressor: exogenously-determined share of landlord bargaining power

Sector heterogeneity matters for agglomeration, but less than expected -
dense firms of all kinds still share knowledge

Other measures of distance beyond spatial matter for agglomeration -
transport links are a good substitute

There are multiple equilibria for agglomeration steady-states:

Landlord bargaining power is a key determinant -
more landlord LVC means less agglomeration, and more misallocation

Field

Macroeconomics: Growth & Productivity

The role of agglomeration in the firm production function, including spillovers

Drawing from **Urban Economics**

Analysis using **Microeconometric** techniques

Local Growth Plans in 12 Combined Authorities in England¹

HMT AoRI²

1.15: What policies and interventions would be most effective in driving regional and local economic growth, and how do these vary across places?

3.2: How can the financial sector best support the real economy, both locally in the places financial firms agglomerate, but also across the country? Are there ways in which the sector's support for the real economy could be improved?

¹English Devolution White Paper

²HM Treasury: Areas of Research Interest

Firm density vs productivity growth (cities, time)

Density vs productivity (global cities)

3-factor production function

$$Y_t = A_t K_t^\alpha L_t^\beta Z_t^\gamma \quad (1)$$

$$Z_t(r) = \int_0^S \psi(r, s) \cdot s \cdot \theta(s) n(s) \, ds \quad (2)$$

$Z_t(r)$: external agglomeration technology

Reduced form estimation

Macroeconomic Policy

Machine Learning in Economics

Time Series Econometrics

Labour Economics

Data (sources)

Source	Granularity	Location	Period
1. FAME	Medium & large firms	HQs only	2006-2025 (20-year rolling)
2. ONS - ABS	Sample of 2000 firms	✓	
3. ONS - all	All firms	✓	
4. HMT - travel times	1km ² gridsquares	Non-firm specific	Instrument (1Y cross-section)

Data (calibration & estimation)